Mengdi Wang

Postdoctoral Fellow, University of British Columbia

Vancouver, BC, Canada mengdi.wang@ubc.ca 6033221138 wang-mengdi.github.io

Research Summary

My research focuses on physical simulation and Physical AI. I have developed multiscale numerical methods and adaptive GPU solvers, and my recent work investigates domain-guided agents for simulation construction and learning physical dynamics and robot policies from simulated experience.

Research Areas: Physical AI, Multiscale and Multiphysics Simulation, Numerical Methods, AI Agents, Computer Graphics

Academic Appointments

University of British Columbia, Postdoctoral Fellow 2026.9 - Present

- Department of Computer Science, PhysAI Lab, working with Prof. Peter Yichen Chen

Education

Georgia Institute of Technology, Ph.D. in Computer Science 2024.1 - 2026.5

- School of Interactive Computing, advised by Prof. Bo Zhu

Dartmouth College, Ph.D. Program in Computer Science (Transferred) 2020.9 - 2023.12

- Guarini School of Graduate and Advanced Studies, advised by Prof. Bo Zhu

Peking University, B.S. in Computer Science, *summa cum laude* 2016.9 - 2020.6

- Center on Frontiers of Computing Studies, advised by Prof. Baoquan Chen

Publications

WorldAgent: Verification-Guided Agentic Physical World Construction Submitted to ICLR 2027

Caoliwen Wang*, *Mengdi Wang*[*†], Yige Chen*, Zejia Wu, Bowen Huang, Siyuan Chen, Guanxiong Chen, Lifu Wei, Heng Zhang, Qinghai Zhang, Yin Yang, Guandao Yang, Shiying Xiong, Peng Wang, Chenfanfu Jiang, Peter Yichen Chen

[*] Co-first authors. [†] Corresponding author.

WorldContact: A Contact-Centric World Model for Scalable Robot Learning Submitted to ICRA 2027

Caoliwen Wang*, *Mengdi Wang*[*†], Heng Zhang, Shixun Huang, Siyuan Chen, Chao Liu, Anpei Chen, Zhendong Wang, Peter Yichen Chen[†], Huamin Wang

[*] Co-first authors. [†] Corresponding authors.

WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer SIGGRAPH Asia 2026

Caoliwen Wang*, Minghao Guo*, Siyuan Chen*, Heng Zhang, *Mengdi Wang*, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, Peter Yichen Chen

[*] Co-first authors.

Matrix-Free Multigrid with Algebraically Consistent Coarsening on Adaptive Octrees Submitted to JCP

Mengdi Wang, Yuchen Sun, Bo Zhu

Cirrus: Adaptive Hybrid Particle-Grid Flow Maps on GPU SIGGRAPH 2025

Mengdi Wang, Fan Feng, Junlin Li, Bo Zhu (Featured in video trailer)

Real-Time Knit Deformation and Rendering SIGGRAPH 2025

Tao Huang*, Haoyang Shi*, *Mengdi Wang**, Yuxing Qiu, Yin Yang, Kui Wu

* Co-first authors.

An Interface Tracking Method with Triangle Edge Cuts JCP 2025

Mengdi Wang, Matthew Cong, Bo Zhu

Hydrophobic and Hydrophilic Solid-Fluid Interaction SIGGRAPH Asia 2022

Jinyuan Liu, *Mengdi Wang*, Fan Feng, Annie Tang, Qiqin Le, Bo Zhu

A Moving Eulerian-Lagrangian Particle Method for Thin Film and Foam Simulation SIGGRAPH 2022

Yitong Deng, *Mengdi Wang*, Xiangxin Kong, Shiying Xiong, Zangyueyang Xian, Bo Zhu

A Clebsch Method for Free-Surface Vortical Flow Simulation SIGGRAPH 2022

Shiying Xiong, Zhecheng Wang, *Mengdi Wang*, Bo Zhu (Featured in video trailer)

Thin-Film Smoothed Particle Hydrodynamics Fluid SIGGRAPH 2021

Mengdi Wang, Yitong Deng, Xiangxin Kong, Aditya H. Prasad, Shiying Xiong, Bo Zhu (Featured in video trailer)

Visual Data Analysis and Simulation Prediction for COVID-19 arXiv preprint (2020)

Baoquan Chen, Mingyi Shi, Xingyu Ni, Liangwang Ruan, Hongda Jiang, Heyuan Yao, *Mengdi Wang*, Zhenhua Song, Qiang Zhou, Tong Ge

Research Experience

Research Intern, Lightspeed Studio, Tencent IEG, 2024.5 - 2024.8 — Los Angeles, CA

- Designed a real-time yarn-level knit fabric simulation framework, resulting in a co-first-author SIGGRAPH 2025 paper.

Research Intern, NVIDIA, 2022.6 - 2022.9 — Santa Clara, CA

- Designed a GPU-accelerated geometric multigrid Poisson solver and VOF-based interface tracking method, published in *Journal of Computational Physics*. Mentored by Ken Museth, Eftychios Sifakis.

Teaching Experience

Guest Lecturer, University of British Columbia — Fall 2026
CPSC 314 Computer Graphics

Instructor, Dartmouth College, COSC 294 (Graduate Seminar) — Fall 2021

Additional Instructor, Dartmouth College / Georgia Tech — Multiple terms

Head Teaching Assistant, Dartmouth College, COSC 277 Computer Graphics — Fall 2023

Contributions to the Research Community

I led the development of MESO, an open-source C++/CUDA simulation framework, independently implemented Cirrus, and have led the continued development and maintenance of SimpleX since 2022. These research codebases have supported publications at SIGGRAPH and NeurIPS and in Journal of Computational Physics.

Peer review: ACM Transactions on Graphics (SIGGRAPH), Computer Graphics Forum (CGF), Journal of Computational Physics